

# Statistical philosophy

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## GOALS

- Discuss what statistics are used for, and why they are needed
- Explain what P values mean, and what they don't
  - Effect sizes and confidence intervals are usually better
- Explain the fundamentals of the two basic paradigms of statistical philosophy
- Discuss the role of statistics in science

## 1 Statistical inference

- We use statistics to confirm effects, estimate parameters, and predict outcomes
- It usually rains when I'm in Cape Town, but mostly on Sunday
  - *Confirmation*: In Cape Town, it rains more on Sundays than other days
  - *Estimation*: In Cape Town, the *odds* of rain on Sunday are 1.6–2.2 times higher than on other days
  - *Prediction*: I am confident that it will rain at least one Sunday the next time I go

## Raining in Cape Town

- How we interpret data like this necessarily depends on assumptions:
  - Is it likely our observations occurred by chance?
  - Is it likely they *didn't*?

## Vitamin A

- We compare health indicators of children treated or not treated with vitamin A supplements
  - *Estimate*: how much taller (or shorter) are the treated children on average?
  - *Confirmation*: are we sure that the supplements are helping (or hurting)?
  - *Range of estimates*: how much do we think the supplement is helping?

## 1.1 P values and confidence intervals

- We use *P values* to say how sure we are that we have seen a positive effect
- We use *confidence intervals* to say what we think is going on (with a certain level of confidence)
- P values are *over-rated*
- *Never* use a high P value as (direct) evidence for anything, e.g.:
  - that an effect is small
  - that two quantities are similar
- What does a P value actually measure?

### Vitamin A example

- We want to know if vitamin A supplements improve the health of village children
  - Is height a good measure of general health?
  - How will we know height differences are due to our treatment?
    - \* **Answer:** We want the two groups to start from the same point – independent randomization of each individual
    - \* **Answer:** We may measure *changes* in height
    - \* **Answer:** Or *control for* height (and maybe other factors)

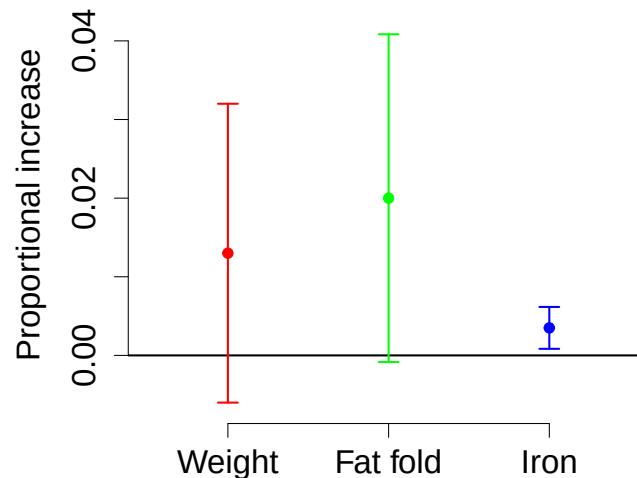
### What do we hope to learn?

- Is vitamin A good for these children?
- How sure are we?
- How good do we think it is?
- How sure are we about that?

### P values

- Activity: What is a P value?
- What does it mean if I find a “significant P value” for some effect in this experiment?
- **Answer:** The difference is unlikely to be due to chance
  - **Answer:** So what! I already know vitamin A has strong effects on metabolism
- If I’m certain that the true answer isn’t exactly zero, why do I want the P value anyway?

## Confidence intervals



- What do these results mean?
- Which are significant?

## Confidence intervals and P values

- A high P value means we can't see the *sign* of the effect clearly
- A low P value means we can
- What would happen if we could measure many more children?

## What do P values measure?

- Answer: Clarity!
- Answer: We should call it that
  - Answer: Instead of significance

## Types of Error

- Type I (*False positive:*) concluding there is an effect when there isn't one
  - This doesn't happen in biology. There is always an effect.
- Type II (*False negative:*) concluding there is no effect when there really is
  - This *should* never happen, because we should never conclude there is no effect

- Type Zero is the error of using numerical codes for things that have perfectly good simple names
- Just say “false positive” or “false negative” when possible

## Experimental design

- *False positive*: in the *hypothetical* case that the effect is exactly zero, what is the probability of falsely finding an effect?
  - Should be less than or equal to my significance value
- *False negative*: what is the probability of failing to find an effect that is there?
  - Requires you specify a hypothetical effect *size*
  - This is a *scientific* judgment
- These are useful to analyze **power** and **validity** of a statistical design
  - You should do these analyses *before* you collect data

## Doing Science

- Once you collect data:
- You should never make a false-negative error.
  - **Answer**: Never accept the null
- Or a false-positive error
  - **Answer**: If you never believe the null is exactly true, the idea of a false positive doesn’t make sense
- *Sign error*: if I think an effect is positive, when it’s really negative (or vice versa)
- *Magnitude error*: if I think an effect is small, when it’s really large (or vice versa)
- Confidence intervals aid clear thinking

## Low P values

- If I have a low P value I can see something clearly
- But it’s usually better to focus on what I see than the P value

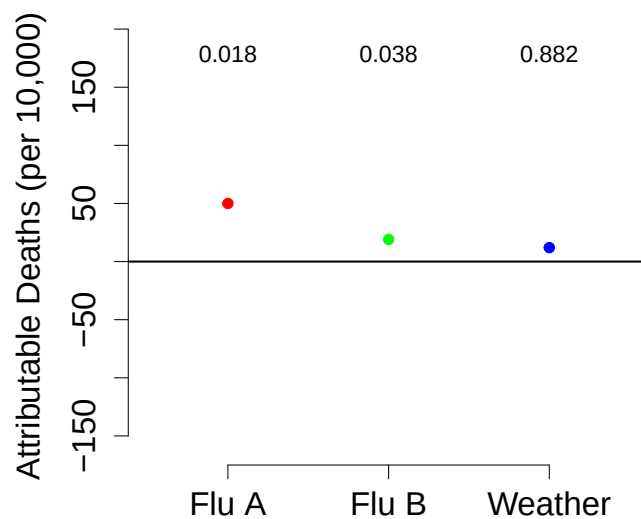
## High P values

- If I have a high P value, there is something I *don’t* see clearly
- It *may be* because this effect is small
- High P values should *not* be used to advance *any* conclusion

## What causes high P values?

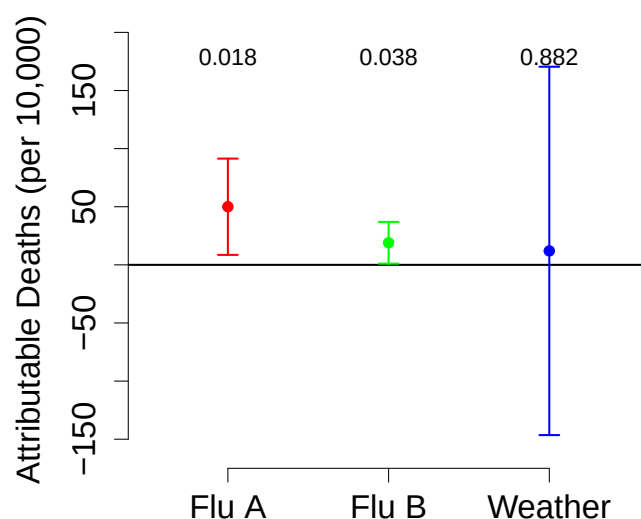
- Small differences
- Less data
- More noise
- An inappropriate model
- Less model resolution
- A lower P value means that your evidence for difference is better
- A higher P value means that your evidence for similarity is better – or worse!

## Annualized flu deaths



- Look at deaths on an annual time scale
- Why is weather not causing deaths at this time scale?
- **Answer:** Maybe those people would have died anyway later in the same winter

... with confidence intervals



- **Never** say: A is significant and B isn't, so  $A > B$
- **Instead:** Construct a statistic for the hypothesis  $A > B$ 
  - May be difficult

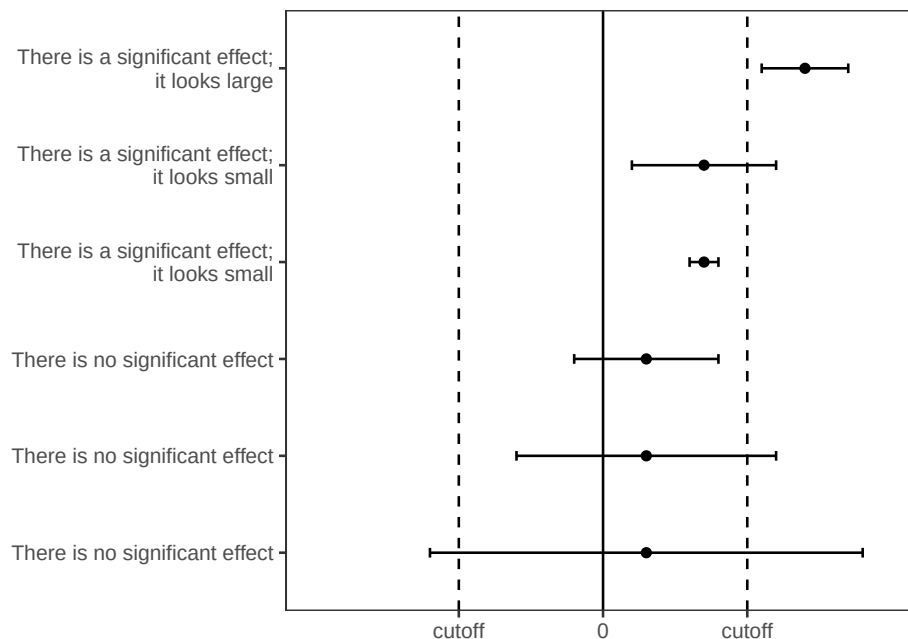
## Misleading language

- Null effects of boot camps
  - **Answer:** Wrong!
- Fat storage in vole populations is not correlated with elevation
  - **Answer:** Yes, it is!
- As expected, the placebo group did not differ significantly from the control group
  - **Answer:** Why would that be good?
- B and B showed that there is no statistically significant difference in sexual risk behaviour between men with and without clinic access in Zambia
  - **Answer:** No, they didn't
  - **Answer:** Statistical significance is a property of the *study* (and the sample population), never of real groups (or the idealized population)

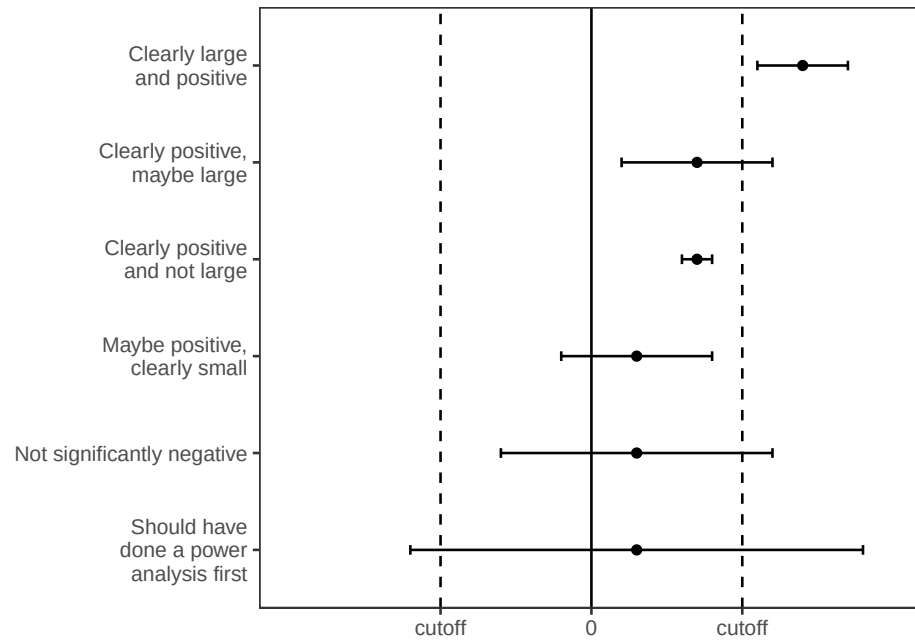
## Another way to talk

- Unclear effects of boot camps
  - **Answer:** They really want to say “small” effects, but that would require a different test!
- As expected, the sign of the difference between the placebo group and controls was unclear
  - **Answer:** Is this convincing? Will it remind people to check the size of difference?
- The direction of correlation between fat storage and elevation in vole populations is unclear in this study
- B and B showed that there is an unclear difference in sexual risk behaviour between men with and without clinic access in Zambia
  - **Answer:** Now obviously weird. So you have to say it better

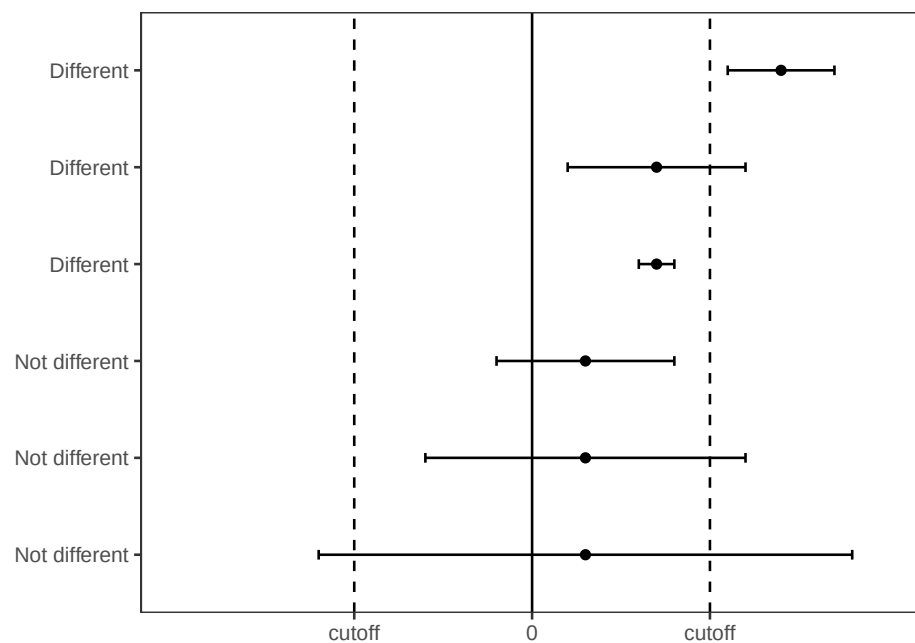
## Confidence intervals



## Confidence intervals



## Confidence intervals



## Cutoffs

- How do I know what is a large effect?
- This can't be done by statistics!
  - Must be scientific
  - It's often hard, and people avoid it



- This leads to sloppy thinking

## 1.2 Statistics and science

### Syllogisms

- All men are mortal
- Mohamed Salah is mortal
- Therefore, Mohamed Salah is a man

### Syllogisms

- All men are mortal
- Fanny the elephant is mortal
- Therefore, Fanny the elephant is a man

### Bad logic

- A lot of statistical practice works this way:
  - bad logic in service of conclusions that are (usually) correct
- This sort of statistical practice leads in the aggregate to bad science
- The logic can be fixed:
  - Estimate a difference, or an interaction

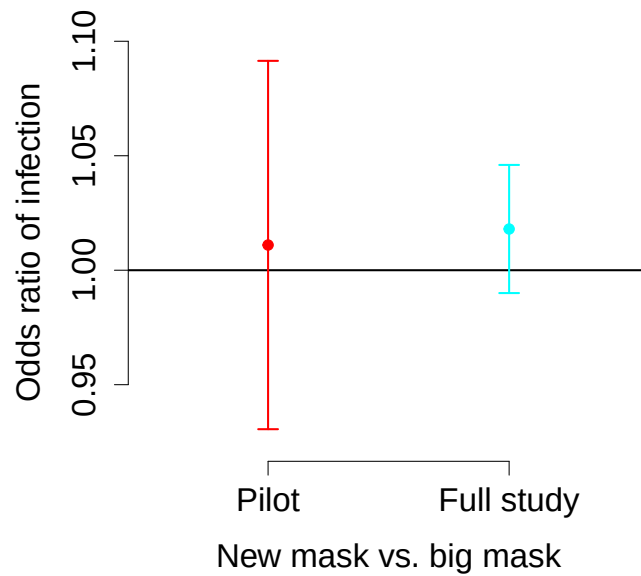
### Small effects

- We can't build statistical confidence that something is small by failing to see it clearly
- We must instead see clearly that it is small
- This means we need a standard (our cutoff) for what we mean by small

### Flu mask example

- People who work in respiratory clinics sometimes have to wear bulky, uncomfortable, expensive masks
- They would like to switch to simpler masks, if those will do the job
- How can this be tested statistically? We don't want the masks to be "different".
  - We need to decide what we mean by different in this case!
  - They're not the same, so how close is close enough?

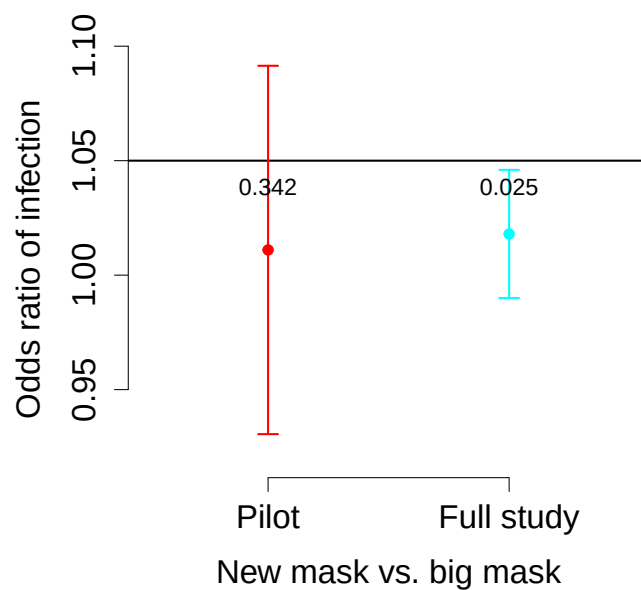
## Traditional statistics



## Non-inferiority approach

- Are we confident the new mask is “good enough”?
- There is no substitute for picking a cutoff
- Can be controversial

## Non-inferiority approach



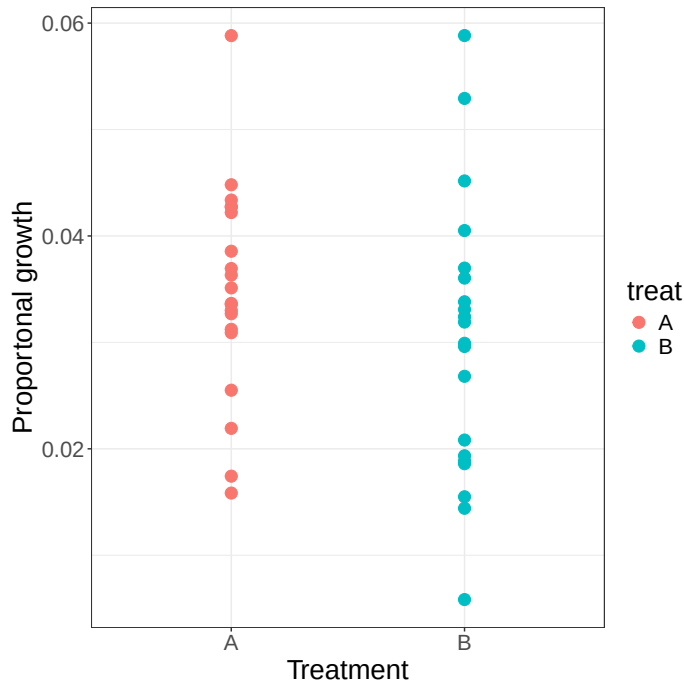
- We can even attach a P value by basing it on the “right” statistic.
- The right statistic is the thing whose sign we want to know:
  - The difference between the observed effect and the standard we chose

## 2 Paradigms for inference

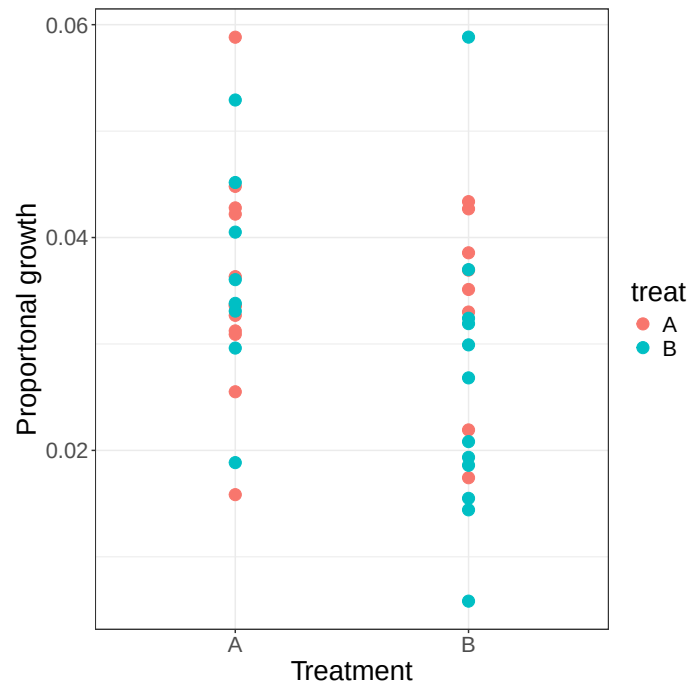
### 2.1 Frequentist paradigm

- Make a null model
- Test whether the effect you see could be due to chance
  - What is the probability of seeing a difference of exactly 0.0048 in proportional growth?
    - \* **Answer:** Zero!
- Test whether the effect you see *or a larger effect* could be due to chance
  - This probability is the (frequentist) P value
- JD: It’s actually better to calculate a one-sided P value. You would then usually double this to get a standard P value

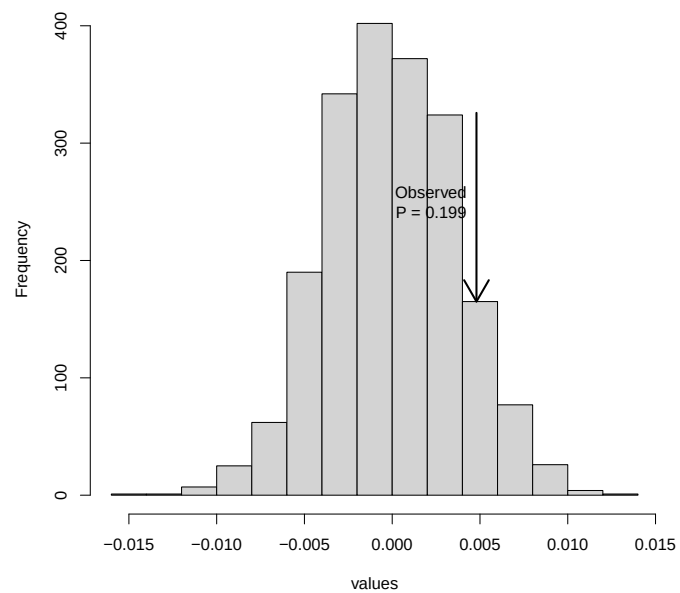
### Height measurements



## Scrambled measurements



## The null distribution



## 2.2 Bayesian paradigm

- Make a complete model world
- Use conditional probability to calculate the probability you want

## A powerful framework

- More assumptions  $\implies$  more power
- With great power comes great responsibility

## Bayesian inference

- We want to go from a *statistical model* of how our data are generated, to a probability model of parameter values
  - Requires *prior* distributions:
    - \* the assumed likelihood of parameters before these observations are made
  - Use Bayes theorem to calculate *posterior* distributions:
    - \* the inferred likelihood of parameters after taking the data into account
- Provides a strong framework for combining information from different sources and for propagating uncertainty

## Bayesian P value

- Once you make strong-enough assumptions, the Bayesian framework allows you to calculate true probabilities
  - Meaning, they are really probabilities; they are only as “true” as your assumptions
- A natural approach to Bayesian P values is simply the probability that your estimate has the wrong sign
  - Often better to double this value, to align with frequentist tradition

## Vitamin A study

- A frequentist can do a clear analysis right away
- A Bayesian needs a ton of assumptions – will often try to make “uninformative” assumptions

## Cape Town weather

- Frequentist: how unlikely is the observation, from a random perspective?
- Bayesian: what’s my model world? What is my prior belief about weather-weekday interactions?

## Example: MMEV

- MMEV is a viral infection that can cause a serious disease (called MMED)
- MMED patients are unable to control their urge to fit models to data
- The rapid MMEV test gives a positive result:
  - 100% of the time for people with the virus
  - 5% of the time for people without the virus

## MMED MMEV questions

- The rapid MMEV test gives a positive result:
  - 100% of the time for people with the virus
  - 5% of the time for people without the virus
- The population prevalence of MMEV is 1%
- You test a *random* person from this population, and the result is positive.
  - What is the probability that they have MMEV?
- This calculation is the core of Bayes *theorem*

## MMED MMEV questions

- You learn that your friend has had a positive rapid test for MMEV
  - What do you tell them?
    - \* **Answer:** You don't tell them *anything* until you ask them some questions!
- Bayesian inference:
  - Clear conclusions
  - based on clear (and often strong) assumptions
  - Need to be clear about the assumptions you are making
- Medical “paradox” video: <https://www.youtube.com/watch?v=lG4VkPoG3ko>

## 3 Philosophy of fitting

- Principled statistical fitting is hard
- A big advantage of having good computers is that you can *test* fitting approaches on simple and complicated model worlds before applying them to real data

## Approximating probabilities

- In both paradigms, we want to calculate true probabilities to reinforce our conclusions
- In most complicated situations (e.g., dynamical model fits), these probabilities cannot be calculated directly
- We approximate the probabilities we want using likelihoods

## Likelihoods

- What we want to know is [frequentist or Bayesian] probabilities relating to our *parameters* based on the data
  - Is  $\mathcal{R}_0 < 2$ ; Is  $\bar{D} < 6d$ ?
- What we can easily calculate based on assumptions is the likelihood of our *data* based on the parameters.
  - Proportional to probability of seeing the data
  - But probability of seeing these exact data is hard to define, and not very interesting

## Algorithms

- If we can calculate likelihoods, we have good algorithms to estimate the probabilities that we want
  - For frequentist tests, we use maximum likelihood estimation
  - For Bayesian tests, we use MCMC

## Normal likelihoods

- Assume that the difference between the estimate  $\hat{y}_i$  and the data point  $y_i$  is normally distributed. What is the log likelihood?

•

$$L = \prod_i \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{-(y_i - \hat{y}_i)^2}{2\sigma^2}\right)$$

•

$$\ell = \sum_i -\log(\sigma\sqrt{2\pi}) - \sum_i \frac{(y_i - \hat{y}_i)^2}{2\sigma^2}$$

- We maximize the likelihood by minimizing the sum of squares
  - and then solving for  $\sigma$

## Least squares → likelihood

- Attaching your least squares fit to a likelihood means:
  - You can *use it* for statistical inference (likelihood ratio test)
  - You can *challenge* the assumptions

## Other distributions

- Maybe we want a lognormal distribution of titres
  - Dr. Rong did that: least squares on the log scale!
- Maybe we want a Poisson (or negative binomial) distribution for counts
- Maybe we want a binomial (or beta binomial) distribution for samples
- It's better to be explicit about what distribution you are assuming!

## 4 Conclusion

### Your philosophy

- Statistics are not a magic machine that gives you the right answer
- If you are to be a serious scientist in a noisy world, you should have your own philosophy of statistics
  - Be pragmatic: your goal is to do science, not get caught by theoretical considerations
  - Be honest: it's harder than it sounds.

### Honesty

- You can always keep analyzing until you find a “significant” result
  - If you do this you will make a lot of mistakes
- You may also keep analyzing until you find a result that you already believe is true.
  - This is *confirmation bias*; you're probably right, but your project is not advancing science
- Good practice
  - Make an analysis plan before you start
  - If flexibility is allowed in your context:
    - \* keep an analysis journal as you go



## Summary

- P values are over-rated
  - and often misunderstand
- High P values should not be used as evidence for anything ever.
  - They can provide indirect evidence. Wonderful. Find the direct evidence and use that instead.
- Effect sizes and confidence intervals are almost always better
- Use the right P value (or confidence interval) for your question
  - Non-inferiority tests, interactions
- Frequentist statistics makes weak assumptions, and finds logically weak formal conclusions:
  - These parameters are unlikely to produce a statistic this extreme by chance
- Bayesian statistics makes strong assumptions:
  - prior distributions must be fully specified
- ...and finds logically strong formal conclusions:
  - The probability that the effect value is in this range is X
  - These strong conclusions can be used directly for prediction with uncertainty
- Statistics are a key component of data-based science
  - You should think about statistical analysis from the beginning of your project
- You need a basic understanding of statistical principles
- You need your own statistical philosophy
  - If you're a theoretician, it should be ideological and honest
  - If you're a scientist, it should be pragmatic and honest